Electronically configurable microscopic metasheet robots

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Nature Materials
DOI: 10.1038/s41563-024-02007-7

Abstract

Shape morphing is vital to locomotion in microscopic organisms but has been challenging to achieve in sub-millimetre robots. By overcoming obstacles associated with miniaturization, we demonstrate microscopic electronically configurable morphing metasheet robots. These metabots expand locally using a kirigami structure spanning five decades in length, from 10 nm electrochemically actuated hinges to 100 μm splaying panels making up the ~1 mm robot. The panels are organized into unit cells that can expand and contract by 40% within 100 ms. These units are tiled to create metasheets with over 200 hinges and independent electronically actuating regions that enable the robot to switch between multiple target geometries with distinct curvature distributions. By electronically actuating independent regions with prescribed phase delays, we generate locomotory gaits. These results advance a metamaterial paradigm for microscopic, continuum, compliant, programmable robots and pave the way to a broad spectrum of applications, including reconfigurable micromachines, tunable optical metasurfaces and miniaturized biomedical devices.